SAP CPI API Management

* What is an API
* Why organizations need APIM
* What SAP API management, how it works
  What it actually does with SAP Cloud Integration

→ What is an API

API stands for Application programming Interface

∴ Set of Rules and protocols that allows one application to communicate with other application

→ Key points : API acts a "messenger" between Application's

(Request → process → Response)

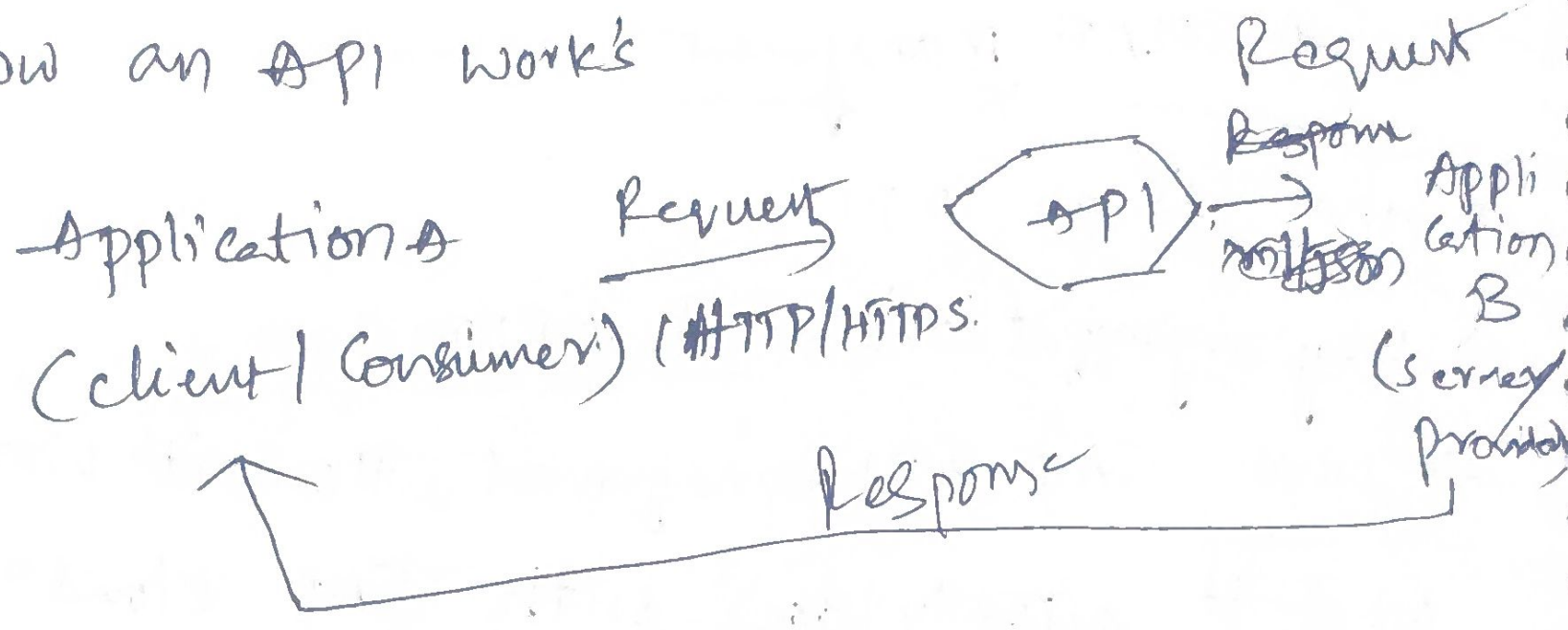
: Enables systems to share data securely between the systems.

: Common protocol : HTTP/ HTTPS

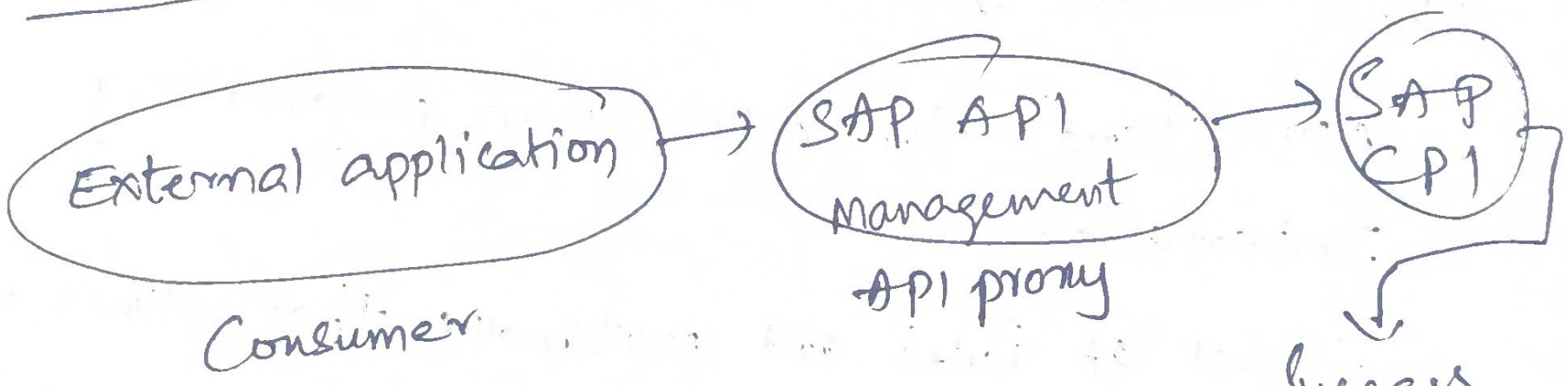
: Common style : Get, post, update, delete
Put

: Returns the data in JSon, XML

* How an API work's



Common Example:



Realtime example:

An external System (or) application Wants Employee data from Success factor's Instead of direct access, it Call's an API

Factors / S4/HANA (Backend System)

Flow: Consumer Sends Request for Employee data → API management Validates and forward's the Request → CPI procey the request Call's the Backend System

Backend System → Response Sent back to the Consumer

* Why Do we need API management.

With out API management



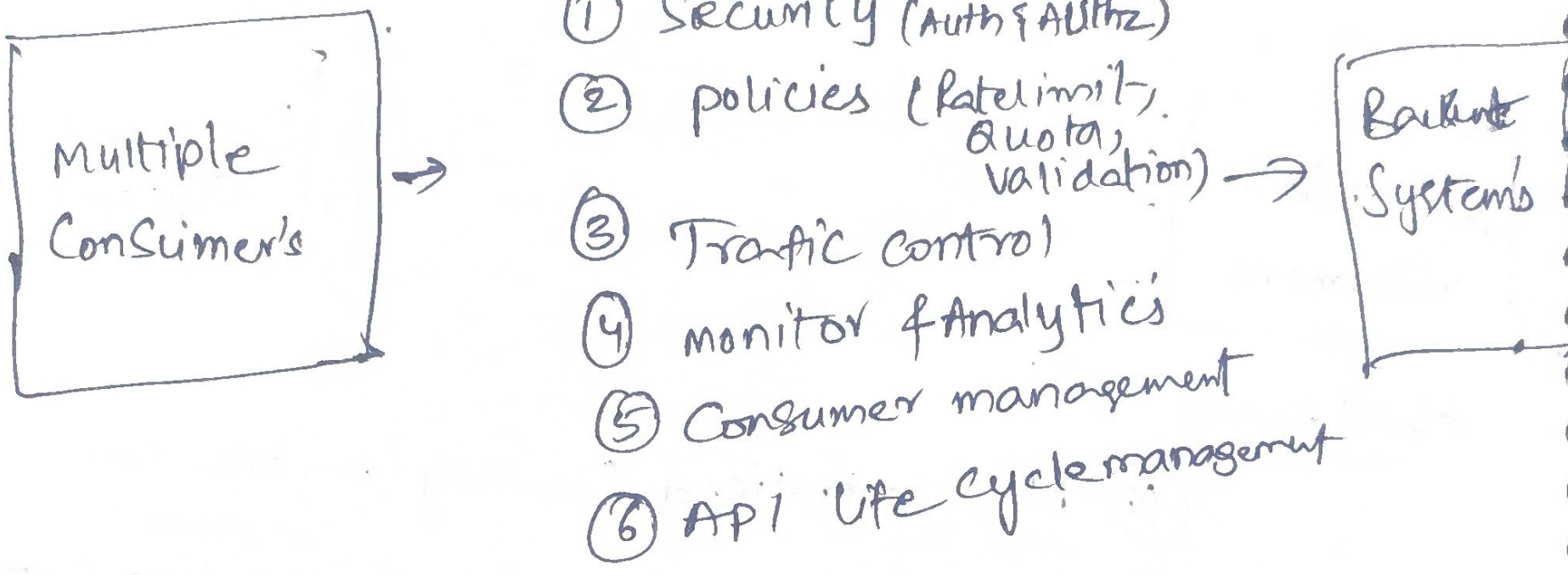
challanges:

- (X) No Centralised Security (Anyone can access)
- (X) No Authentication (or) Authorization Control
- (X) No rate limiting (Risk of System overload)
- (X) No visibility. (Hard to monitor API usage)
- (X) No Standardization & governance
- (X) High maintainance & governance
- (X) Security vulnerabilities and Compliance Risk.

* With API management

It's a centralized location

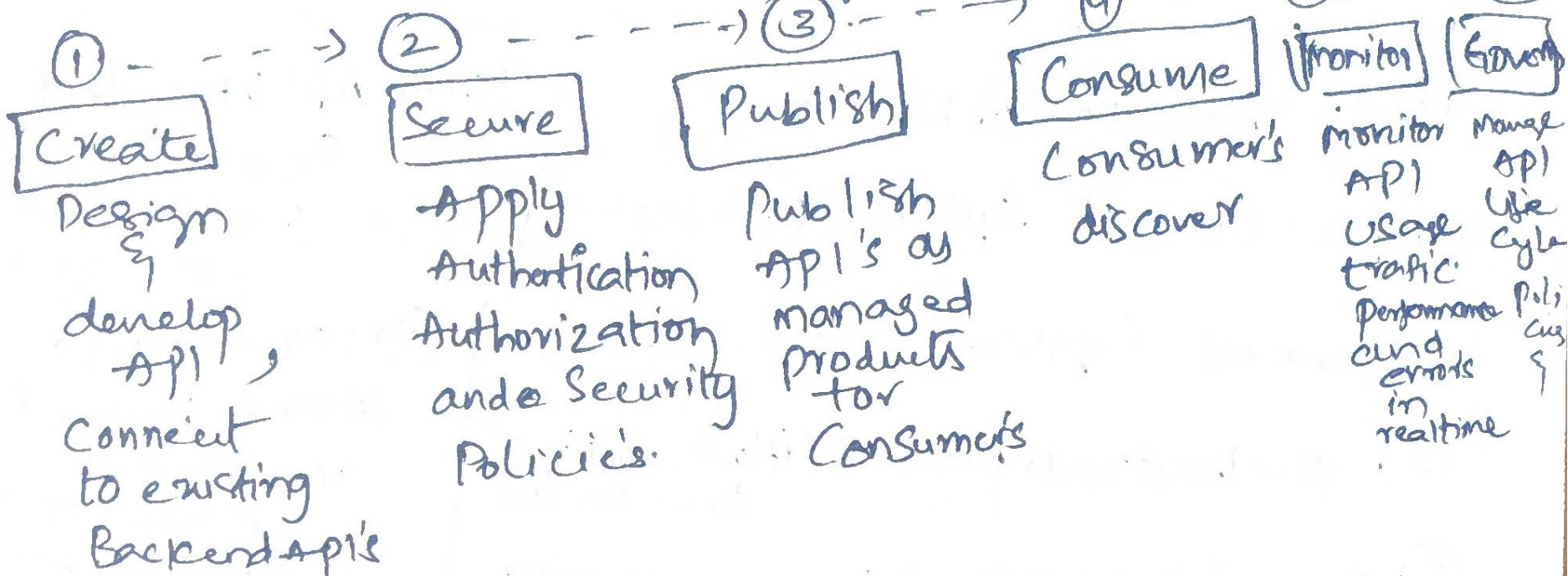
With API Management



* Bottom line: API management act's as a Smart gateway between Consumer's and backend System's ensuring Security Control, visibility and great experience of both business and developer's

* What is API management
  → It is not simply exposing API
  * Complete life cycle it will manage from create to governance

The API mangement lifecycle



* API management in SAP Integration Suite

→ SAP API Management is a capability of SAP Integration Suite used to publish, Secure, monitor and manage API's

Key objects of API management

(x) already discussed these points above.

* NOW we are going to discuss about the CPI VS API management — What's the difference

→ CPI and API management are Complementary Capabilities in SAP Integration Suite, They Serve different purpose but work together in most enterprise Scenarios

Sap cloud Integration

Focus: Integration / Orchestration

purpose: Connect Systems & processes

① Connect Systems (cloud, onpremise)

② orchestrate process (iflows, routing, transformation)

③ Transform & map (message mapping, XSLT, Groovy, Content enrichment)

④ Adapters (prebuilt Adapters S4 HANA, HTTP, JDBC, JMS, SFTP, IDoc, mail, SOAP, oData)

⑤ Business Logic (using scripting, routers, content modifiers)

⑥ Scheduling (scheduling, Batch processing, and monitoring of integrations)

Sap API management

Focus: API lifecycle & Governance.

purpose: Expose, Secure & manage

① Expose APIs (Expose backend Service and Integration) flows as APIs

② Secure API's (Authentication, Authorization, OAuth 2.0, API keys, Certificate

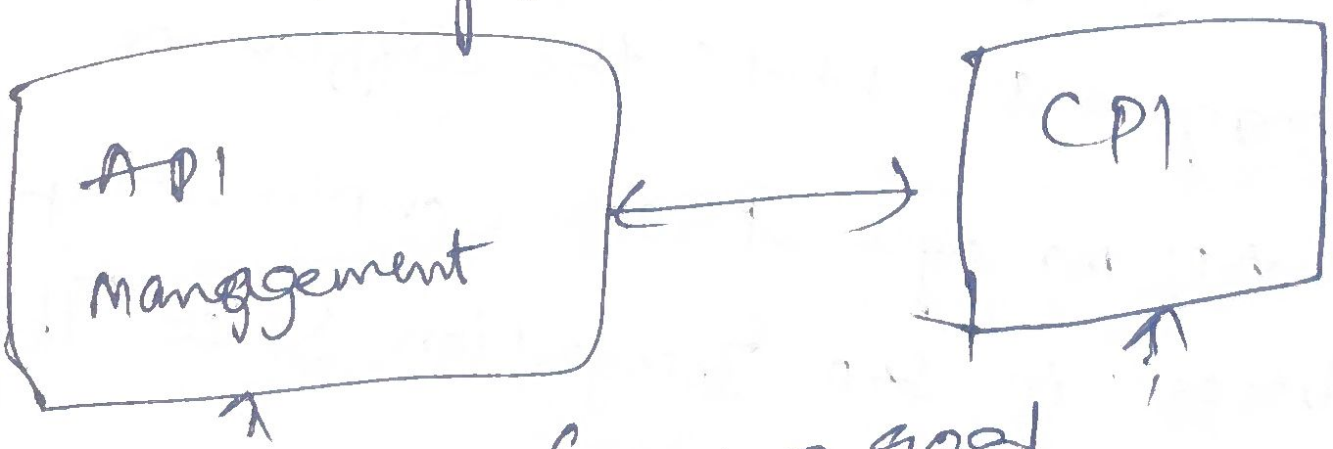
③ Apply policies (Rate limiting, Quotas, (Throttling, Transformation validation)

④ Manage Consumers

⑤ monitor & Analyze (monitor API Traffic performance errors)

⑥ API lifecycle (API versions, lifecycle, portals)

They work Together



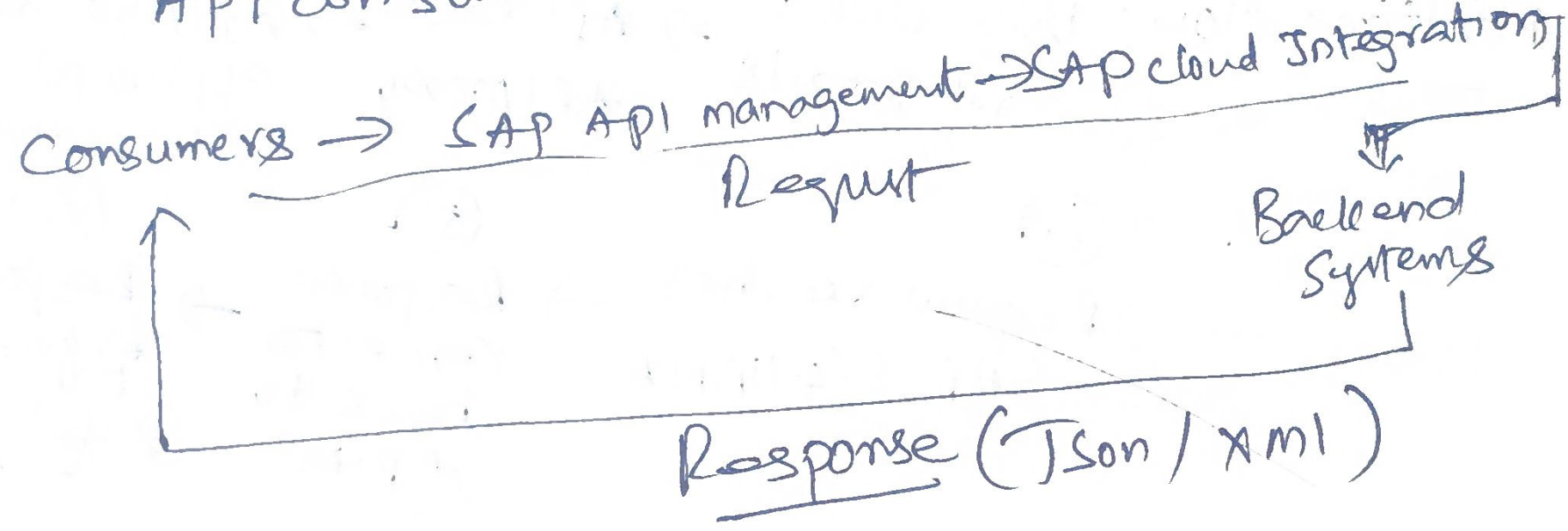
Common goal

Deliver Secure, reliable, and Scalable Integrations and API's

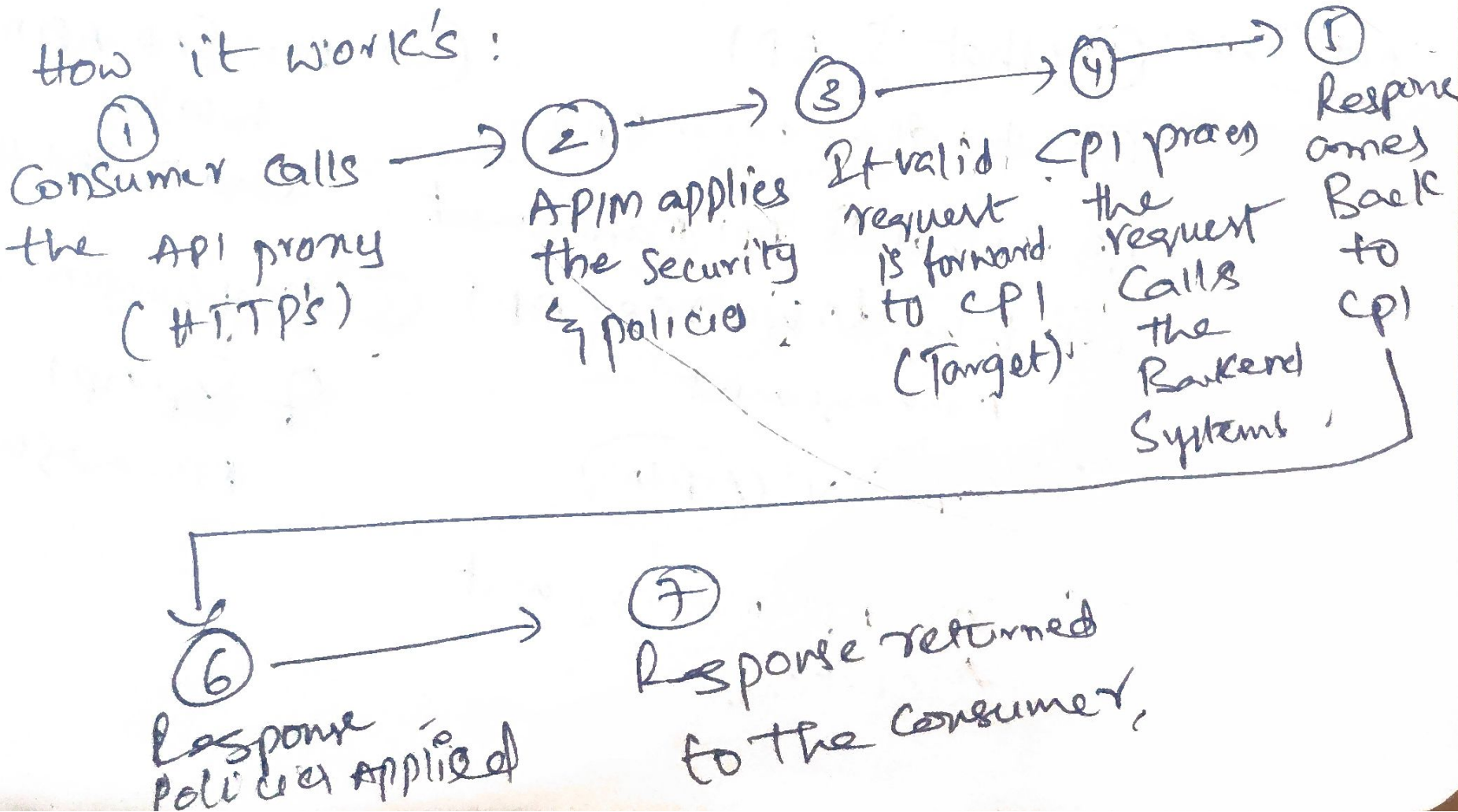
* Use API management to manage, secure and govern your API
* Use CPI to build the Integration logic and connect to backend Systems

SAP API management Architecture

→ API management → Secure gateway → ..
API consumers → backend System



How it works:



Real time example:

A mobile app request's Sales order details from SAP S/4 HANA in real time SAP API Management -



Realtime flow: (1) User clicks Get Details → (2) API calls API proxy → (3) API validates applies policies routes request → (4) Request reaches SAP S/4 HANA → (5) Response comes back to APIM → (6) Response Returned to Mobile App

Outcome:
1. What is API
2. Why do we need API
3. What is Api management
4. Why do we need API management
5. CPI VS APIM
6. SAP Api management Architecture
7. How SAP APIM Works End to End flow
8. Key components of SAP API management